

Section 7

Systems Programming ECE3502

SIC-XE DISASSEMBLER

HPRG1X00100000001D

T0010001DB410B400B44075101000E32019332FFADB2013A00433200857C003B850
E001000

Given the following necessary information:

AND

Assume **B=101A**

NOTE THAT:

This HTME is not covering the whole program

Instruction	Opcode	Format
CLEAR r1	B4	2
LDT	74	3/4
TD	E0	3/4
RD	D8	3/4
JEQ	30	3/4
COMPR r1,r2	A0	2
STCH	54	3/4
TIXR r1	B8	2

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

1st instruction:
B410B4

Op code	ni xbpe	Disp
1011 01	00	

Since n=0 & i=0, then the instruction is
neither format 3 nor format 4

B4 op code => CLEAR (format 2)

1st instruction:
B410

Op code	R1	R2
1011 0100	0001	0000

CLEAR

X

H**PROG1X001000**00001D
T**001000**1DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000				

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

2nd instruction:
B400B4

Op code	ni xbpe	Disp
1011 01	00	

Since n=0 & i=0, then the instruction is
neither format 3 nor format 4

B4 op code => CLEAR (format 2)

2nd instruction:
B400

Op code	R1	R2
1011 0100	0000	0000

CLEAR A

H**PROG1X001000**00001D
T**001000**1D**B410**B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002				

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

3rd instruction:
B44075

Op code	ni xbpe	Disp
1011 01	00	

Since n=0 & i=0, then the instruction is
neither format 3 nor format 4

B4 op code => CLEAR (format 2)

3rd instruction:
B440

Op code	R1	R2
1011 0100	0100	0000

CLEAR S

H**PROG1**X00100000001D
T0010001D**B410**B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004				

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

4th instruction:
751010

Op code	ni xbpe	Disp
0111 01	01 0001	hexa (01000)

Since n=0 & i=1, then **immediate #**
Since e=1 => **format 4 +**

4th instruction:
75101000

Op code => return discarded bits

Op code	ni xbpe	Address
0111 0100	01 0001	hexa (01000)

74 op code => LDT

+LDT #4096

Address
value in
decimal

H**PROG1**X**001000**00001D
T**001000**1D**B410****B400****B440**75101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006				

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

5th instruction:
E32019

Op code	ni xbpe	Disp
1110 00	11 0010	hexa (019)

Since n=1 & i=1, then **direct**
Since e=0 => **format 3**
Since b=0 & p=1, then **PC relative**

TA= disp+PC = 019+100D = 1026
Op code => return discarded bits

Op code	ni xbpe	Disp
1110 0000	11 0010	hexa (019)

E0 op code => TD

TD Ref1

H**PROG1**X**001000**00001D
T**001000**1DB**410**B**400**B**440**7**5101000**E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A				

Symbol	Address
Ref1	1026

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

6th instruction:
332FFA

Op code	ni xbpe	Disp
0011 00	11 0010	hexa (FFA)

Since n=1 & i=1, then **direct**
Since e=0 => **format 3**
Since b=0 & p=1, then **PC relative**

TA= disp+PC = FFFFFFFA+1010 = 100A
Op code => return discarded bits

Op code	ni xbpe	Disp
0011 0000	11 0010	hexa (FFA)

30 op code => JEQ

JEQ Ref2

H**PROG1**X00100000001D
T0010001DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A		TD	Ref1	E32019
100D				

Symbol	Address
Ref1	1026
Ref2	100A

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

7th instruction:
DB2013

Op code	ni xbpe	Disp
1101 10	11 0010	hexa (013)

Since n=1 & i=1, then **direct**
Since e=0 => **format 3**
Since b=0 & p=1, then **PC relative**

TA= disp+PC = 013+1013 = 1026
Op code => return discarded bits

Op code	ni xbpe	Disp
1101 1000	11 0010	hexa (013)

D8 op code => RD

RD Ref1

H**PROG1**X00100000001D
T0010001DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A	Ref2	TD	Ref1	E32019
100D		JEQ	Ref2	332FFA
1010				

Symbol	Address
Ref1	1026
Ref2	100A

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

8th instruction:
A00433

Op code	ni xbpe	Disp
1010 00	00	

Since n=0 & i=0, then the instruction is
neither format 3 nor format 4

A0 op code => COMPR (format 2)
8th instruction:
A004

Op code	R1	R2
1010 000	0000	0100

COMPR A , S

H**PROG1**X00100000001D
T0010001DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A	Ref2	TD	Ref1	E32019
100D		JEQ	Ref2	332FFA
1010		RD	Ref1	DB2013
1013				

Symbol	Address
Ref1	1026
Ref2	100A

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

9th instruction:
332008

Op code	ni xbpe	Disp
0011 00	11 0010	hexa (008)

Since n=1 & i=1, then **direct**
Since e=0 => **format 3**
Since b=0 & p=1, then **PC relative**

TA= disp+PC = 008+1018 = 1020
Op code => return discarded bits

Op code	ni xbpe	Disp
0011 0000	11 0010	hexa (008)

30 op code => JEQ

JEQ Ref3

H**PROG1X001000**00001D
T**001000**1DB**410B400B44075101000E32019332FFAD**B**2013A004**33200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A	Ref2	TD	Ref1	E32019
100D		JEQ	Ref2	332FFA
1010		RD	Ref1	DB2013
1013		COMPR	A,S	A004
1015				

Symbol	Address
Ref1	1026
Ref2	100A
Ref3	1020

Assume that the instruction is format 3
so we are going to take 6 hexadecimal

10th instruction:
57C003

Op code	ni xbpe	Disp
0101 01	11 1100	hexa (003)

Since n=1 & i=1, then **direct**
Since e=0 => **format 3**
Since x=1 => **indexed ,X**
Since b=1 & p=0, then **Base relative**

TA= disp+B = 003+101A = 101D
Op code => return discarded bits

Op code	ni xbpe	Disp
0101 0100	11 1100	hexa (003)

54 op code => STCH

STCH Ref4

H**PROG1X001000**00001D
T**001000**1DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A	Ref2	TD	Ref1	E32019
100D		JEQ	Ref2	332FFA
1010		RD	Ref1	DB2013
1013		COMPR	A,S	A004
1015		JEQ	Ref3	332008
1018				

Symbol	Address
Ref1	1026
Ref2	100A
Ref3	1020
Ref4	101D

11th instruction:
B850
B8 op code => TIXR (format 2)

Op code	R1	R2
1011 1000	0101	0000

TIXR T

HPROG1X00100000001D
T0010001DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A	Ref2	TD	Ref1	E32019
100D		JEQ	Ref2	332FFA
1010		RD	Ref1	DB2013
1013		COMPR	A,S	A004
1015		JEQ	Ref3	332008
1018		STCH	Ref4,X	57C003
101B				

Symbol	Address
Ref1	1026
Ref2	100A
Ref3	1020
Ref4	101D

H**PROG1**X00100000001D
T0010001DB410B400B44075101000E32019332FFADB2013A00433200857C0
03B850
E001000

	PROG1	Start	1000	-----
1000		CLEAR	X	B410
1002		CLEAR	A	B400
1004		CLEAR	S	B440
1006		+LDT	#4096	75101000
100A	Ref2	TD	Ref1	E32019
100D		JEQ	Ref2	332FFA
1010		RD	Ref1	DB2013
1013		COMPR	A,S	A004
1015		JEQ	Ref3	332008
1018		STCH	Ref4,X	57C003
101B		TIXR	T	B850
101D	Ref4			

Symbol	Address
Ref1	1026
Ref2	100A
Ref3	1020
Ref4	101D